Victor Crespo-Rodriguez

Melbourne, AU | +61 483 020 645
me@victorcrespo.xyz |linkedin.com/in/crespovictor | victorcrespo.xyz

Research Profile

Final-year PhD student in Information Technology with a focus on software testing for autonomous systems. My research involved building test processes and methodologies for generation of safety-critical situations from the ground up, bridging software testing methodologies and high-level AI decision-making. I am passionate about solving complex systems-level problems and am eager to apply my skills in software testing, AI and robotics frameworks, and functional safety principles.

Technical Skills

- **Programming Languages:** Python, Ruby, Bash
- **Automotive Frameworks and Systems:** BeamNg Simulator, CARLA Simulator, Apollo Platform, ISO 26262
- **Systems Software:** Linux, Real-Time Operating Systems (RTOS), QEMU
- **Tools & Methods:** Git, Docker, MATLAB

Research & Project Experience

PhD Researcher | Faculty of Information Technology – Monash University | Melbourne | 2021 – 2025
Thesis Topic: "Search-Based Software Testing for Autonomous Vehicles"

- **Designed and implemented** a methodology for creating safety-critical situations for an autonomous vehicle, adhering to software testing guidelines, and implemented it on an Autonomous Vehicle simulation platform.
- **Implemented a custom methodology** based on a MATLAB toolkit for assessing software testing techniques to answer the question *"What makes a test case an effective one?"*
- **Created a modular testing system** for improving the creation of effective test suites that generated new and diverse test scenarios for autonomous systems. Implemented a Machine Learning based approach for predicting outcomes, eliminating the need for running simulations.
- **Collaborated closely** with a team of Professors, Doctors, and PhD students specializing in software testing of AI and real-time autonomous systems to optimize software platforms and methods for performance and power efficiency.

Technical Support Engineer – Onfleet | USA/Remote | 2021 – 2023

- **Provided technical support** to clients of the Onfleet last-mile delivery platform in APAC Geography.
- **Onboarded and guided clients** in their integrations to the platform by creating tutorials and knowledge base articles integrated into the Onboarding Guidelines.

Technical Support Lead – Arcus Financial Intelligence | Mexico | 2016 – 2020

- **Led the Tech Support Team**, providing Level 2 technical support, best practices, and leading conversations with stakeholders and clients for integrating them to the Arcus Platform and API.
- **Created and optimised** technical support workflows and processes in the Support Team, including Technical and Customer Support platforms. Created and updated knowledge bases for internal use. tutorial on creating ISO 26262-inspired safety monitors for ROS 2 nodes, which was featured on the ROS Discourse forum.

Support Engineer – Microsoft | Mexico | 2014 – 2016

- **Provided technical support** to clients of the Microsoft Dynamics suite in Latin America. Guided clients in their integrations and acted as single point of contact for clients' queries.
- **Part of the MACH (Microsoft Academy for College Hires),** a graduate program for building leadership skills with hands-on experience. Started the GLEAM (LGBT Employees at Microsoft) Employee Resource Group in Mexico and helped to create other Latin American chapters.

Education

Ph.D. in Information Technology | Monash University | Melbourne | 2025
Thesis Topic: "Search-Based Software Testing for Autonomous Vehicles "
Advisor: Professor Aldeida Aleti

Master of Applied Cybernetics, *with Commendation* | The Australian National University | Canberra | 2021

B.S. in Mechatronics Engineering, *graduated with Honourable Mention* | Tec de Monterrey | Mexico | 2014

Publications & Presentations

- **Crespo-Rodriguez, Victor**, Neelofar, and Aldeida Aleti. "PAFOT: A Position-Based Approach for Finding Optimal Tests of Autonomous Vehicles." *Proceedings of the 5th ACM/IEEE International Conference on Automation of Software Test (AST 2024)*. 2024.
- **Crespo-Rodriguez, Victor**, et al. "Instance Space Analysis of Testing of Autonomous Vehicles in Critical Scenarios." *ACM Transactions on Software Engineering and Methodology* 34.3 (2025): 1-36.
- **Crespo Rodriguez, Victor**, et al. "The Role of Road Features and Vehicle Dynamics in Cost-Effective Autonomous Vehicles Testing: Insights from Instance Space Analysis." *Pre-print Available at SSRN 5314027*.